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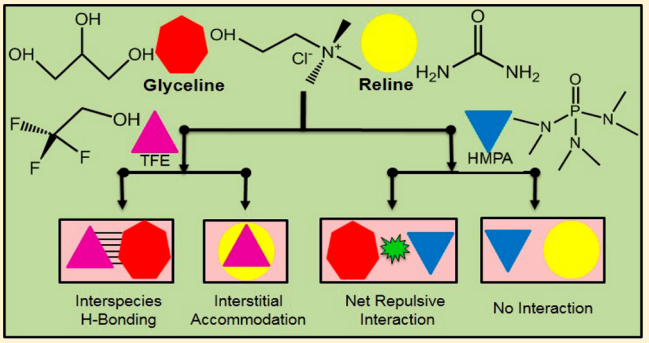
Hydrogen Bond Donor/Acceptor Cosolvent-Modified Choline Chloride-Based Deep Eutectic Solvents

Ashish Pandey, Bhawna, Divya Dhingra, and Siddharth Pandey*

Department of Chemistry, Indian Institute of Technology Delhi, Hauz Khas, New Delhi 110016, India

Supporting Information

ABSTRACT: Deep eutectic solvents (DESs) have emerged as nontoxic and inexpensive alternatives not only to the common organic solvents but to the ionic liquids as well. Some of the common and popular, and perhaps the most investigated, DESs are the ones comprising an ammonium salt and an appropriate hydrogen bond (HB) donor in a predetermined mole ratio. The formation of the DES is attributed to the H-bonding interaction(s) present between the salt and the HB donor. Consequently, addition of a predominantly HB donor or a predominantly HB acceptor cosolvent to such DESs may result in intriguing features and properties. We present investigation of two DESs constituted of salt choline chloride along with HB donors urea and glycerol, respectively, in 1:2 mol ratio, named reline and glyceline as the cosolvent of very high HB donating acidity and no HB accepting basicity 2,2,2-trifluoroethanol (TFE) and of very high HB accepting basicity and no HB donating acidity hexamethylphosphoramide (HMPA), respectively, is added. TFE shows up to 0.25 mole fraction miscibility with both reline and glyceline. While up to 0.25 mole fraction HMPA in glyceline results in transparent mixtures, this cosolvent is found to be completely immiscible with reline. From the perspective of the solvatochromic absorbance and fluorescence probes, it is established that the cybotactic region dipolarity within up to 0.25 mole fraction TFE/HMPA-added DES strongly depends on the functionalities present on the solute. Fourier transform infrared absorbance and Raman spectroscopic investigations reveal no major shifts in vibrational transitions as TFE/HMPA is added to the DES; spectral band broadening, albeit small, is observed nonetheless. Excess molar volumes and excess logarithmic viscosities of the mixtures indicate that while TFE may interstitially accommodate itself within H-bonded network of reline, it does appear to form H-bonds with the constituents of the glyceline. Increase in overall net repulsive interactions as HMPA is added to glyceline is suggested by both positive excess molar volumes and excess logarithmic viscosities. The addition of HB donor/acceptor cosolvent appears to disturb the salt−HB donor equilibria within DES via complex interplay of interactions within the system.



INTRODUCTION

Deep eutectic solvents (DESs) have emerged in past decade or so as inexpensive and potentially environmentally benign alternatives to ionic liquids.[1−5] Some of the common and popular DESs consist of a salt and a hydrogen bond (HB) donor.[1−5] Among salts, a quaternary ammonium salt, choline chloride (ChCl), has been used extensively to prepare DESs as it is inexpensive (ChCl is an important additive in chicken feed).[2−7] To prepare a ChCl-based DES, a HB donor is simply mixed with ChCl in predetermined proportions and heated gently to form a liquid. Such ChCl-based DESs have been used in many applications in chemistry and chemical technology; the most notable being biodiesel preparation,[8] biotransformations,[9,10] polymer synthesis,[11] catalysis,[12] and nanoscience[13] among many others.

Although applications of DESs as solvent media in various chemical processes have started to emerge, their potential use in chemical applications is often hindered by their limited and, in some cases, undesirable physicochemical properties (for example, most ChCl-based DESs exhibit relatively higher viscosities). Properties of the DESs may be tuned by changing salt/HB donor combination, but the extent of change is often fairly limited. Subsequently, cosolvent-modified DESs are becoming a topic of interest and curiosity as they have the potential to exhibit significantly altered properties.[14−18] In this context, judicious selection of a cosolvent is the key to affording the desired system properties. Further, as the H-bonding between the anion of the salt (i.e., Cl^- in ChCl) and the HB donor is proposed to be the major reason for the lowered freezing point of the DES when the salt and the HB donor are mixed in right proportion,[1−7] we envisage that a cosolvent having either high HB donating (HBD) acidity or high HB accepting (HBA) basicity may be able to significantly alter the properties of the DES.

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Toward this, we have selected 2,2,2-trifluoroethanol (TFE), a fluorinated solvent that demonstrates excellent environmental and biological compatibility,[19−21] as a cosolvent which can be added to a DES to alter its properties. The main reason to select TFE is that, as a solvent, it has very high HBD acidity and almost negligible HBA basicity.[21,22] Because of the electronegativity of the trifluoromethyl group, TFE, as a solvent, exhibits significantly stronger HBD acidity as compared to conventional short-chain alcohols (e.g., ethanol) and is known to form stable H-bonded complexes with heterocycles, such as THF and pyridine.[23−26] Quantitatively, in terms of Kamlet−Taft empirical parameters, the HBD acidity parameter, α, of TFE is ∼1.00 and the HBA basicity parameter, β, is ∼0.[21,22] It is hypothesized that due to high α value, TFE may compete with the HB donor constituent of the DES toward H-bonding with ChCl thus causing structural changes within the system. On similar lines, hexamethylphosphoramide (HMPA), a polar-aprotic solvent, is selected as a cosolvent, because its HBD acidity, α, is ∼0, whereas it has a very high HBA basicity, β, of ∼1.00. HMPA is a colorless mobile liquid which is miscible with water and many polar and nonpolar organic solvents, but is usually immiscible with saturated hydrocarbons.[27] The symmetrical charge distribution over the $N_3P^+-O^-$ grouping and the high electron density on the oxygen give rise to a large dipole moment and unusually high basicity.[27] HMPA is shown to have applications in the synthesis of polymers, gases, and organometallic compounds.[27] Because of the lone pair availability on the three dimethyl-substituted nitrogens along with the lone pairs on oxygen bonded to phosphorus, there is a possibility that HMPA may compete with ChCl in H-bonding interactions with the HB donor of the DES.

To assess the role of the structure of the HB donor of the DES on interactions with added cosolvents TFE and HMPA, respectively, the two DESs selected for investigation are constituted of ChCl salt and urea (named reline) and ChCl salt and glycerol (named glyceline) in 1:2 mol ratio. While the HBD ability of glycerol is due to the three alcoholic−OH groups; that of urea is through the hydrogens on the two nitrogens of the amide functionality. To gauge the changes in the interactions as HBD cosolvent TFE or HBA cosolvent HMPA is added to the DES reline or glyceline, we have used a set of absorbance and fluorescence spectroscopic probes. On the basis of the spectroscopic probe responses, we have assessed empirical solvent parameters depicting polarity, HBD acidity, and HBA basicity of the TFE/HMPA-containing reline/glyceline mixtures. The outcomes of these experiments are corroborated by noninvasive measurements of FTIR absorbance and Raman spectra of the mixtures along with densities and dynamic viscosities under ambient conditions. Our results unveil interesting trends in interactions present within the TFE- and HMPA-added reline/glyceline-based mixed solvent systems. These interactions appear to impart modified physicochemical properties to these (and related) cosolvent systems, further expanding the application potential of DES-based media in chemical analysis.

EXPERIMENTAL SECTION

Materials. 2,6-Dichloro-4-(2,4,6-triphenyl-*N*-pyridino)-phenolate (betaine dye 33), 4-nitroaniline (NA), and *N*,*N*-diethyl-4-nitroaniline (DENA) were purchased in the highest available purity from Fluka (≥99%, HPLC), Spectrochem. Co. Ltd., and Frinton Laboratories, respectively, and recrystallized multiple times before use. The fluorescence probes pyrene (Py) [≥99.0% (GC), *puriss* for fluorescence], 1-anilino-8-naphthalenesulfonate (ANS) [99%], and 6-propionyl-2-(dimethylamino)naphthalene (PRODAN) [≥98% (HPLC)] were obtained in the highest purities from Sigma-Aldrich, Acros Organics, and Biochemika, respectively, and were used as received. Both DESs, glyceline and reline, were purchased in highest purity from Scionix Ltd. and were stored in inert environment before their use. Alternatively, glyceline and reline were prepared by mixing ChCl (≥99% from Sigma-Aldrich) with glycerol (≥99.5%, spectrophotometric grade from Sigma-Aldrich) and urea (≥99% from Sigma-Aldrich), respectively, in a mole ratio of 1:2 followed by stirring under heating at ∼80 °C until a homogeneous, colorless liquid has been formed. All spectroscopic measurements on DESs purchased from Scionix Ltd. and those prepared by mixing ChCl with the corresponding H-bond donor were found to be statistically the same. Cosolvents TFE (≥99.0%) and HMPA (99%) were purchased from Sigma-Aldrich and were used as received. Absolute ethanol was used to prepare probe stock solutions. Doubly distilled deionized water with ≥18.0 MΩ·cm resistivity was obtained from Millipore Milli-Q Academic water purification system. Structures of DESs used along with cosolvents selected are provided in Figure 1, and structures of the absorbance and fluorescence probes are given in Figure 2.

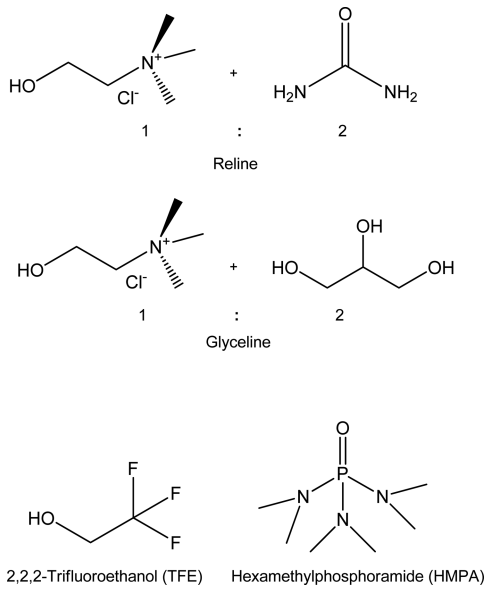


Figure 1. Structures of DESs and HBD/HBA cosolvents used in this study.

Methods. All absorbance and fluorescence probe stock solutions were prepared in ethanol and stored in precleaned amber glass vials at 4 ± 1 °C. The required amount of probes was weighed using a Mettler-Toledo AB104-S balance with a precision of ±0.1 mg. To prepare samples for measurement, an appropriate volume of the probe stock solution was transferred to a clean quartz cuvette and ethanol was evaporated using a gentle flow of high purity nitrogen gas. Neat DES, TFE, or HMPA, and (DES + TFE/HMPA) mixtures of desired compositions prepared by mass, were then introduced to the cuvette to achieve the final desired probe concentration. The solubility of a probe within neat DES, TFE, or HMPA, and (DES + TFE/HMPA) mixture is checked using the linearity of

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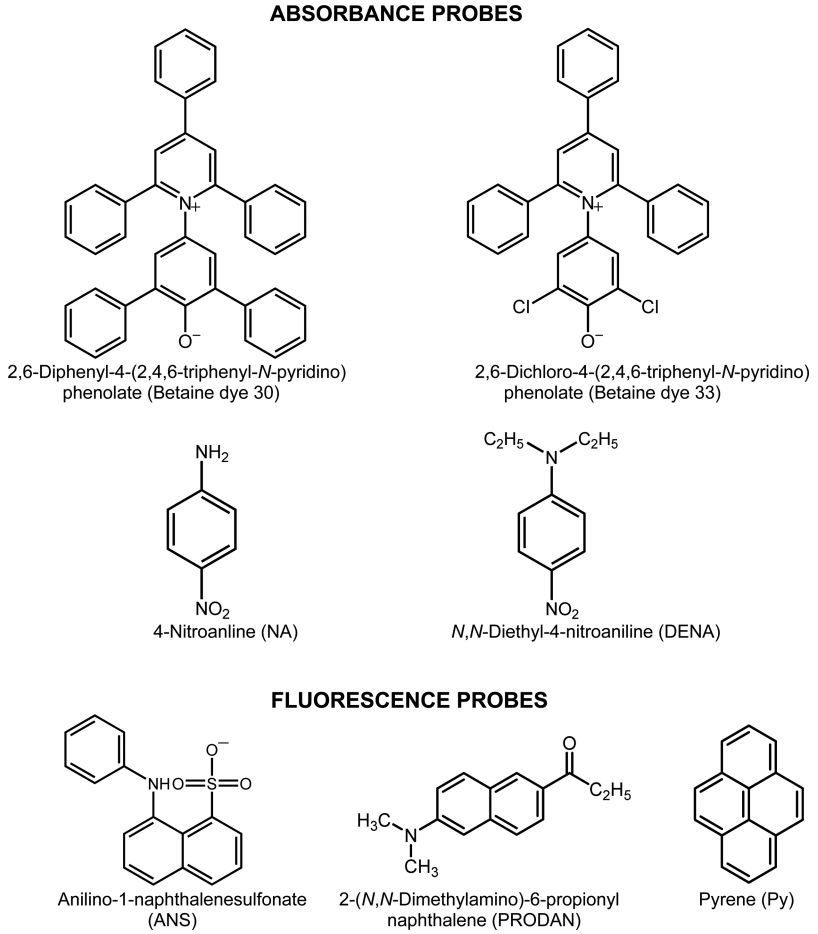


Figure 2. Chemical structures of the polarity-responsive probes used in this study.

the absorbance and/or the fluorescence intensity versus the concentration plot(s).

UV−vis absorbance data were acquired using a PerkinElmer Lambda 35 UV−vis double beam spectrophotometer with variable bandwidth. Horiba-Jobin Yvon Fluorolog-3 (model FL 3-11) modular spectrofluorometer equipped with single Czerny−Turner grating excitation and emission monochromators as wavelength selection devices, a 450 W Xe-arc lamp as the excitation source, and a photomultiplier tube as the detector was used for acquisition of steady-state fluorescence emission spectra. Fluorescence spectra of the probes were collected with the following excitation/emission slit widths (in nm): ANS, 2/2; PRODAN, 1/1; Py, 1/1 under excitation at 346, 350, and 337 nm, respectively. All absorbance and fluorescence data were acquired using 1 cm^2 quartz cuvettes. Attenuated and reflectance-Fourier-transform infrared (ATR-FTIR) absorbance data from 4000 to 400 cm^{-1} were acquired on an Agilent Technologies Cary 660 ATR double-beam spectrophotometer. Raman spectra were acquired with 532 nm excitation using XploRA Confocal Raman (model no. X/01/220) spectrometer. A Peltier-based (resolution of 0.01 °C and accuracy < 0.05 °C) automated Anton Paar microviscometer and a Peltier-based Mettler Toledo DE45 delta range density meter were used to measure dynamic viscosities (η) and densities (ρ) of the (DES + TFE/HMPA) mixtures, respectively. All reported spectroscopic values were averages based on performing triplicate measurements on independently prepared samples. All spectra were duly corrected by measuring the spectral responses from appropriate blanks before data analysis and statistical treatment in each case. All fluorescence probes used were found to have adequate fluorescence quantum yields within DESs and (DES + TFE/HMPA) mixtures under investigation. The errors in ρ and η are less than or equal to ±0.00005 g·cm^{-3} and less than or equal to ±0.5%, respectively.

RESULTS AND DISCUSSION

As the mole ratio of the salt to HB donor decides the extent of lowering of the freezing point of a DES, the addition of an external HB donor/acceptor cosolvent to a DES may result in changes in the freezing points of the DESs reline and glyceline. We noticed that under ambient conditions, while up to 0.25 mole fraction TFE (x_{TFE} = 0.25) addition to reline and glyceline, respectively, showed complete miscibility and rendered clear liquid mixtures, for x_{TFE} = 0.30, the (reline + TFE) sample turned opaque, whereas the (glyceline + TFE) mixture featured the presence of two immiscible liquids (Figure S1). As a result, data collection was only carried out until x_{TFE} = 0.25. We believe that TFE, due to its significantly strong HBD acidity, preferentially starts to H-bond with ChCl in reline resulting in separation of solid urea in the system. In glyceline, the preferential H-bonding of TFE with ChCl renders free glycerol which results in the appearance of two immiscible liquids in the system. For the (glyceline + HMPA) mixture, the sample became opaque for x_{HMPA} = 0.30 (Figure S1). However, for the (reline + HMPA) system, the mixtures became opaque for as small as x_{HMPA} = 0.05 (Figure S1). Therefore, the (reline

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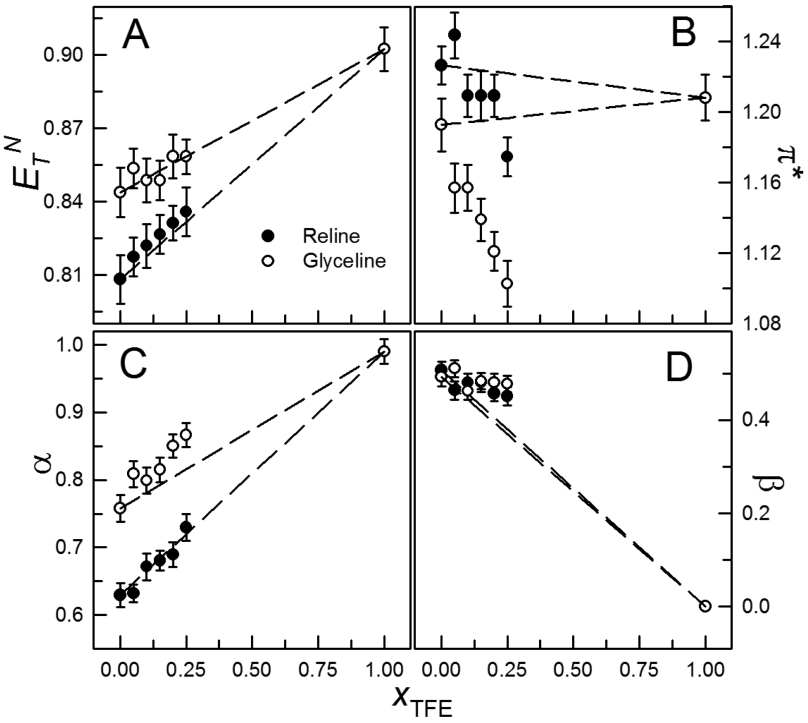


Figure 3. E_T^N (panel A) and Kamlet–Taft parameters–dipolarity/polarizability (π^*, panel B), HBD acidity (α, panel C), and HBA basicity (β, panel D) for TFE-added reline/glyceline mixtures under ambient conditions. Dashed lines represent mole fraction-weighted ideal additive values.

+ HMPA) system could not be investigated any further. As HMPA possesses significantly high HBA basicity, strong H-bonding interactions between nitrogen/oxygen lone pairs of HMPA and hydrogens of $-NH_2$ on urea are tentatively proposed to be reason for this observation. If urea preferentially H-bonds with HMPA instead of ChCl, it would free solid ChCl salt in the mixture. It is to be noted that both ChCl and urea are not soluble in HMPA under ambient conditions.

Behavior of Betaine Dye 33 and Kamlet–Taft Parameters. Betaine dye 30 (or Reichardt's dye), a solvatochromic absorbance probe, exhibits a pronounced band shift in response to changes in solvent polarity (more precisely, dipolarity/polarizability and H-bond donating acidity).[28] The lowest-energy intramolecular charge-transfer (ICT) absorption band of Betaine dye 30 is hypsochromically shifted by ca. 357 nm in going from the nonpolar solvent diphenyl ether (λ_{max}^{abs} = 810 nm) to the highly polar, protic solvent water (λ_{max}^{abs} = 453 nm), where λ_{max}^{abs} is the wavelength of maximal absorption for the ICT band.[28,29] Betaine dye 30 is strongly affected by both the dipolarity/polarizability (π^* parameter) and the HBD acidity (α parameter) of the solvent. The well-known empirical scale of solvent polarity, $E_T(30)$, is defined as the molar transition energy of the dye, traditionally expressed in kcal·mol^{-1}, at standard temperature and pressure according to the expression $E_T(30) = 28591.5/\lambda_{max}^{abs}$ (nm). Because of the solubility restrictions of Betaine dye 30 in DESs, we have instead used Betaine dye 33 in our present investigation. The lowest energy absorbance transition of this dye [$E_T(33)$] is calculated the same way as $E_T(30)$ is calculated. The $E_T(33)$ values can be subsequently expressed in normalized E_T^N values using eqs 1 and 2.[22,28,29]

$$E_T(30) = 0.9953(\pm 0.0287)E_T(33) - 8.1132(\pm 1.6546) \quad (1)$$

$$R = 0.9926, \text{ standard error of estimate} = 0.8320, \ n = 20$$

$$E_T^N = \frac{[E_T(30)_{solvent} - E_T(30)_{TMS}]}{[E_T(30)_{water} - E_T(30)_{TMS}]} \quad (2)$$

where TMS is tetramethylsilane. Using $E_T(30)_{water}$ = 63.1 kcal mol^{-1} and $E_T(30)_{TMS}$ = 30.7 kcal mol^{-1}, eq 2 can be recast as

$$E_T^N = \frac{[E_T(30)_{solvent} - 30.7]}{32.4} \quad (3)$$

E_T^N is thus dimensionless and varies between 0 for TMS (a very nonpolar solvent) and 1 for water (very polar). As mentioned earlier, both dipolarity/polarizability (π^*) and the HBD acidity (α) contribute toward E_T^N. These two empirical solvent parameters along with HBA basicity (β parameter), collectively called Kamlet–Taft solvatochromic parameters,[30−33] were assessed independently for the reline/glyceline mixtures with TFE and HMPA, respectively. The π^* values were estimated from the absorption maxima of DENA ($\bar{\nu}_{DENA}$ in kK), a non-HBD solute, using

$$\pi^* = 8.649 - 0.314\bar{\nu}_{DENA} \quad (4)$$

The α values were determined from $E_T(30)$ and π^* using[34]

$$\alpha = 0.0649E_T(30) - 2.03 - 0.72\pi^* \quad (5)$$

and β values are estimated from the enhanced solvatochromic shift of NA relative to its homomorph DENA by

$$\beta = -0.357\bar{\nu}_{NA} - 1.176\pi^* + 11.12 \quad (6)$$

TFE-Added Reline/Glyceline Mixtures. UV–vis absorbance spectra of Reichardt's dye dissolved in TFE-added reline and glyceline, respectively, were acquired until x_{TFE} = 0.25, and the corresponding E_T^N values were estimated and summarized in Figure 3A. The E_T^N for neat reline/glyceline and that for TFE are in agreement with those reported in the literature earlier.[17,21,35] As expected, E_T^N in neat TFE is significantly

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higher than in neat reline/glyceline due to the higher HBD acidity of TFE. Consequently, we observe that as TFE up to x_{TFE} = 0.25 is added to reline and glyceline, respectively, the E_T^N increases. This increase in E_T^N with x_{TFE} is close to linear suggesting mole fraction weighted ideal-additive behavior of Betaine dye 33 within (reline/glyceline + TFE) mixtures. Because of the higher HBD acidity of glycerol over urea,[35] E_T^N of neat glyceline is higher than that of neat reline. As a result, increase in E_T^N of reline upon TFE addition is more dramatic (higher slope) than that in glyceline. It is important to highlight the differences between E_T^N of TFE-added DESs reline/glyceline and that of TFE-added IL 1-butyl-3-methylimidazolium hexafluorophosphate ([bmim][PF_6]). E_T^N of [bmim][PF_6] increases significantly as TFE is added and becomes closer to the E_T^N of neat TFE even at x_{TFE} = 0.25.[21] The behavior of Betaine dye 33 in TFE-added DESs reline/glyceline is clearly different from that in TFE-added IL [bmim][PF_6]. We believe increase in HBD acidity of the reline/glyceline upon TFE addition is compensated by the decrease in the dipolarity/polarizability of the medium, thus giving rise to expected mole fraction weighted ideal additive behavior.

To explore the validity of this proposition, the π^* (dipolarity/polarizability) values of TFE-added reline/glyceline were estimated from the UV−vis absorbance spectra of DENA using eq 4 (Figure 3B). The π^* of neat TFE is between those of neat reline and neat glyceline (with π^* of reline being a little higher than that of glyceline). It is interesting to note that as TFE up to x_{TFE} = 0.25 is added to reline/glyceline, the π^*, in general, decreases with the decrease being more pronounced for glyceline as compared to that for reline. It is to be noted that a similar decrease in π^* is also observed when TFE is added to [bmim][PF_6].[21] The HBD acidity (α) (estimated from eq 5) of TFE-added reline/glyceline further supports the above outcomes (Figure 3C). As expected, α of neat glyceline is higher than that of neat reline due to the higher HBD acidity of glycerol over urea. The addition of TFE to reline and glyceline, respectively, up to x_{TFE} = 0.25 results in a close to linear increase in α of the mixture. While this linearity is more acceptable for TFE-added reline, the α of the (glyceline + TFE) mixture, in general, is slightly higher than that predicted by the ideal additive behavior (i.e., linearity). In TFE-added glyceline, the α above ideal additive values is compensated by the π^* that are below the ideal additive values resulting in close to linear behavior of E_T^N. This compensation is less pronounced in TFE-added reline mixtures. Finally, as TFE has no HBA basicity ($\beta \approx 0$) and reline/glyceline has similar HBA basicity (Cl^- and −O− of ChCl are responsible for β of these DESs), the addition of TFE to reline/glyceline should result in a decrease in β of the mixtures. The experimental β values of TFE-added reline/glyceline were obtained using eq 6 and are plotted along with the ideal additive values in Figure 3D. In general, a marginal decrease in β is observed as TFE up to x_{TFE} = 0.25 is added to reline and glyceline, respectively. The β of the (reline/glyceline + TFE) are more than those expected from mole fraction weighted ideal additive behavior. This suggests restricted participation of the cholinium −O− in H-bonding interaction with HB donor TFE within the mixture. As the HBA species (ChCl) is the same in both the mixtures, the β of the two mixtures are also statistically the same at each composition investigated.

HMPA-Added Glyceline Mixtures. E_T^N of HMPA-added glyceline (until x_{HMPA} = 0.25) are shown in Figure 4A. As HMPA does not possess appreciable HBD acidity, E_T^N in neat

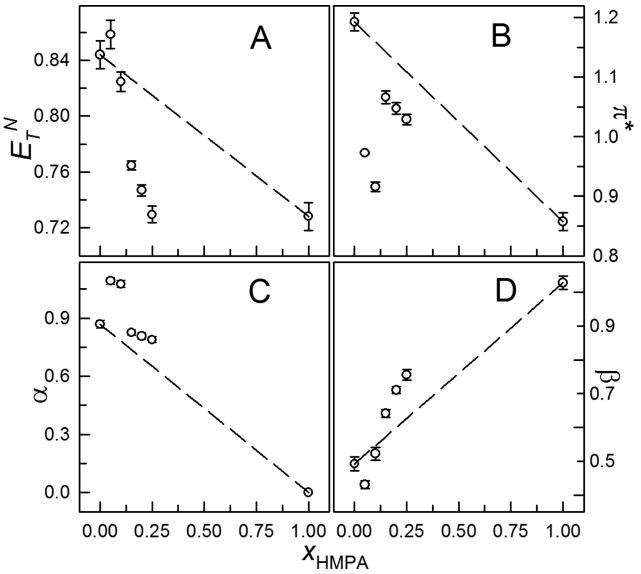


Figure 4. E_T^N (panel A) and Kamlet−Taft parameters−dipolarity/polarizability (π^*, panel B), HBD acidity (α, panel C), and HBA basicity (β, panel D) for HMPA-added glyceline mixtures under ambient conditions. Dashed lines represent mole fraction-weighted ideal additive values.

HMPA is considerably smaller than that in neat glyceline. Consequently, addition of HMPA to glyceline, in general, results in decrease in E_T^N. However, this decrease is fairly dramatic, and E_T^N of (glyceline + HMPA) mixtures are significantly lower than those predicted from mole fraction weighted ideal additive values. The significantly lower than expected E_T^N of HMPA-added glyceline can be explained either by the presence of specific solute−solvent interactions (preferential solvation of Betaine dye 33 by HMPA over glyceline) or solvent−solvent interactions (between HMPA and glyceline). To further pinpoint the reason for this outcome, dipolarity/polarizability, π^*, which is obtained independently of E_T^N, is estimated next and presented in Figure 4B. The π^* is observed to be much higher for neat glyceline than that for neat HMPA. Similar to that observed for E_T^N, as HMPA up to x_{HMPA} = 0.25 is added to glyceline, the decrease in π^* is more than that expected from ideal behavior. However, for the same addition of HMPA (a non-HBD cosolvent) to glyceline (appreciable HBD ability), the decrease in HBD acidity (α, Figure 4C) of the mixture is not more than that expected from ideal additive behavior; instead the α values even increase for the first two additions of HMPA (x_{HMPA} = 0.05 and 0.10) to glyceline before decreasing. The α values of (glyceline + HMPA) mixtures are higher than those predicted from ideal behavior. This rules out the possibility of preferential solvation of the probes Betaine dye 33 and DENA by HMPA. As expected, the HBA basicity (β) of neat HMPA is much higher than that of neat glyceline (Figure 4D), and as HMPA is added to glyceline, the β of the mixture, in general, increases; the values being usually higher than those expected from ideal additive behavior. The E_T^N and the Kamlet−Taft empirical parameters (π^*, α, β) of H-bond acceptor cosolvent HMPA-added DES glyceline hint at the possible presence of solvent−solvent interactions within the mixture.

Behavior of Fluorescence Probes ANS, PRODAN, and Pyrene. To gain deeper insight into the interactions present within reline/glyceline mixtures with TFE and HMPA,

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respectively, we carried out further spectroscopic studies with three well-known fluorescent dipolarity probes: ANS, PRODAN, and pyrene (Py). ANS, a negatively charged probe, undergoes a large variation in its fluorescence responses with solvent polarity due to the intramolecular charge transfer process (ICT).[35−37] However, specific solute−solvent interactions, change in molecular conformation, intersystem crossing to the triplet state and monophotonic photoionization might also contribute to the change in fluorescence responses of ANS with change in solvent dipolarity. It is well-established, nonetheless, that fluorescence emission maxima (λ_{max}^{em}) of ANS shifts bathochromically as the polarity of its cybotactic region is increased.[36,37] Similarly, the λ_{max}^{em} of PRODAN (a neutral ICT probe) is also found to be highly sensitive to solvent polarity, ranging from ca. 392 nm in cyclohexane to 523 nm in water.[38−40] In this state, the carbonyl group intramolecularly gains electron density at the expense of the dimethylamino group, resulting in an increased molecular dipole moment. The pyrene solvent polarity scale (Py I_1/I_3) is defined by the fluorescence intensity ratio of the first and third emission bands for pyrene, where I_1 is the intensity of the solvent-sensitive band arising from the $S_1(v = 0) \rightarrow S_0(v = 0)$ transition and I_3 corresponds to the solvent-insensitive $S_1(v = 0) \rightarrow S_0(v = 1)$ emissive transition.[41,42] The Py I_1/I_3 ratio increases with increasing solvent dipolarity and is a function of both the solvent dielectric (ε) and the refractive index (n) via the dielectric cross term, $f(\varepsilon,n^2)$.[41]

TFE-Added Reline/Glyceline Mixtures. Figure 5A presents ICT λ_{max}^{em} of ANS in TFE-added reline/glyceline mixtures. ANS λ_{max}^{em} in neat TFE is considerably higher than that in neat reline/glyceline suggesting a higher dipolarity of ANS cybotactic region in TFE as compared to that in reline/glyceline. However, as the TFE up to x_{TFE} = 0.25 is added to reline/glyceline, instead of increasing steadily (as suggested by mole fraction weighted ideal additive behavior), the ANS λ_{max}^{em} remains statistically the same. The preferential solvation of charged probe ANS by the DES constituted of ionic species may be tentatively proposed for this outcome. To gain further insight into the interactions present within TFE-added DES mixtures, λ_{max}^{em} of PRODAN, an uncharged ICT probe, is acquired (Figure 5B) from its fluorescence emission spectra in (reline/glyceline + TFE) mixtures. Again, the cybotactic region dipolarity of PRODAN in neat TFE is observed to be higher than that in neat reline/glyceline as PRODAN λ_{max}^{em} in neat TFE is significantly higher than that in neat reline/glyceline. The addition of TFE up to x_{TFE} = 0.15 to reline results in an initial increase in PRODAN λ_{max}^{em}; however, for x_{TFE} = 0.20 and 0.25, the PRODAN λ_{max}^{em} does not increase any further. For TFE-added glyceline mixtures of same x_{TFE}, PRODAN λ_{max}^{em} does not increase and is in fact decreased to some extent. The structure of the cybotactic region of the ICT fluorescence probes appears to depend on the identity of the probe as well as the identity of DES forming the (DES + TFE) mixture.

Pyrene probe response, Py I_1/I_3, and its mole fraction weighted ideal additive values are presented in Figure 5C. As reported previously, Py I_1/I_3 in neat DESs reline and glyceline are significantly high;[35] and they are higher than Py I_1/I_3 in neat TFE. It is to be noted that, unlike ANS and PRODAN, pyrene is devoid of functionality. As expected, the addition of TFE up to x_{TFE} = 0.25 to reline does result in a decrease in Py I_1/I_3; the decrease is more than that expected from ideal additive behavior. On the other hand, the addition of the same concentration of TFE to glyceline shows almost no decrease in Py I_1/I_3, in general. It appears that solvent−solvent interactions between the DES and TFE resulting in different structures of the cybotactic regions may account for such pyrene probe behavior in the (DES + TFE) mixtures.

An overall assessment of the fluorescence polarity probe data (Figure 5) reveals interesting features regarding (reline/glyceline + TFE) mixtures. It appears that irrespective of the fluorescence probe response in neat components constituting the mixture, the responses in (reline/glyceline + TFE) mixtures tend to be similar. This leads us to propose that either the Cl^- H-bonded to TFE or cholinium ion or both control the excited-state cybotactic region of the fluorescence probes within the mixture with the role of DES HBD component being minimal. It is also mention-worthy that the fluorescence probe responses in TFE-added reline/glyceline are fairly different from those observed in ([bmim][PF_6] + TFE) mixtures.[21]

HMPA-Added Glyceline Mixtures. As expected, ANS λ_{max}^{em} of neat HMPA is significantly lower than that of neat glyceline (Figure 6A). The addition of HMPA up to x_{HMPA} = 0.25 results in a decrease in ANS λ_{max}^{em}; however, the decrease is considerably more than that expected from ideal behavior. On similar lines, the PRODAN λ_{max}^{em} is also much lower in neat HMPA as compared to that in neat glyceline; and the decrease in PRODAN λ_{max}^{em} as HMPA is added to glyceline is more than that expected from ideal additive behavior (Figure 6B). The Py I_1/I_3 in neat HMPA is found to be only slightly lower than that in neat glyceline suggesting similar dipolarities of the pyrene cybotactic regions of the two solvents (Figure 6C). However, the addition of HMPA up to x_{HMPA} = 0.25 results in a dramatic decrease in Py I_1/I_3 suggesting a considerable decrease in the cybotactic region dipolarity of the excited pyrene within the

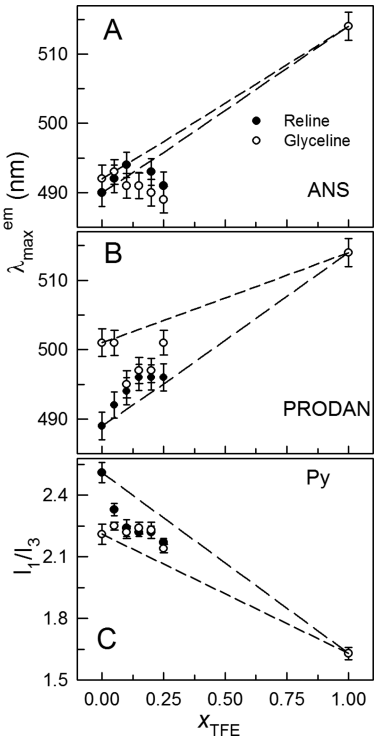


Figure 5. Variation in ANS λ_{max}^{em} (10 μM, λ_{ex} = 346 nm, panel A), PRODAN λ_{max}^{em} (1 μM, λ_{ex} = 350 nm, panel B) and Py I_1/I_3 (1 μM, λ_{ex} = 337 nm, panel C) for TFE-added reline/glyceline mixtures under ambient conditions. Dashed profiles display mole fraction-weighted ideal additive values.

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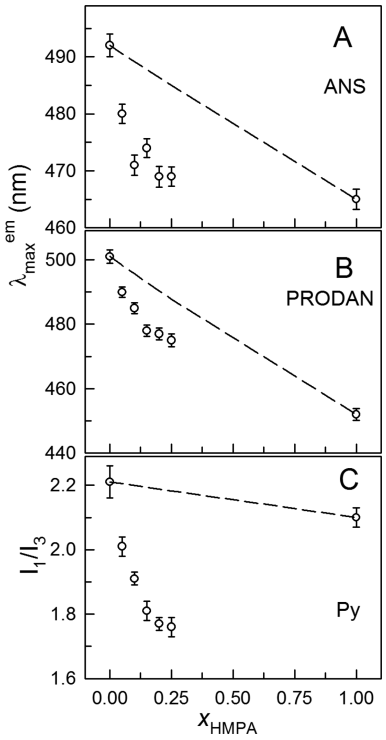


Figure 6. Variation in ANS λ_{max}^{em} (10 μM, λ_{ex} = 346 nm, panel A); PRODAN λ_{max}^{em} (1 μM, λ_{ex} = 350 nm, panel B) and Py I_1/I_3 (1 μM, λ_{ex} = 337 nm, panel C) for HMPA-added glyceline mixtures under ambient conditions. Dashed profiles display mole fraction-weighted ideal additive values.

mixture. It is difficult to explain this decrease in Py I_1/I_3 on the basis of any solute−solvent interactions; solvent−solvent interactions appear to be responsible for these observations.

The overall response of the three fluorescence probes (charged and neutral ICT probes along with a vibronic ratiometric probe) clearly indicate cybotactic regions with lower dipolarity than that expected for HMPA-added glyceline mixtures. It may be inferred that perhaps the HMPA-glycerol (HBA-HBD) H-bonded species, having lowered dipolarity, primarily constitutes the cybotactic region of the fluorescence probe in the (glyceline + HMPA) mixture with the role of ChCl being minimal.

FTIR Absorbance and Raman Spectroscopic Measurements. To further analyze interactions present within reline/glyceline mixtures with TFE/HMPA in a nonintrusive manner (i.e., in the absence of an extrinsic molecular probe), we collected FTIR absorbance and Raman spectra of the mixtures at representative compositions ranging from neat reline/glyceline to neat TFE/HMPA until $x_{TFE/HMPA}$ = 0.25.

TFE-Added Reline/Glyceline Mixtures. The FTIR spectra are collected for neat TFE along with neat reline and (reline + TFE) mixtures (Figure 7A), and neat glyceline and (glyceline + TFE) mixtures (Figure 7B), respectively, until x_{TFE} = 0.25. A careful examination of the data presented in Figure 7 reveals interesting outcomes. The peaks for vibrations associated with the O−H group at 3363 and 3633 cm^{-1} for TFE undergo changes as TFE is added to reline/glyceline. The O−H stretch for "introverted" monomeric TFE, intramolecularly H-bonded between the OH group and fluorine of the CF_3 group that appears at the highest energy (i.e., the band at 3633 cm^{-1}),[43−45] completely vanishes in the presence of reline/glyceline

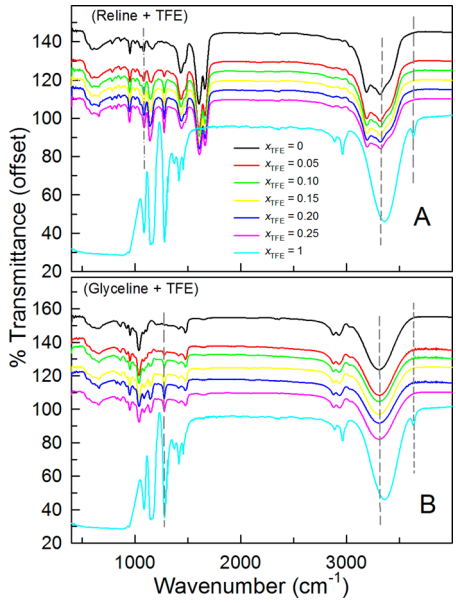


Figure 7. FTIR absorbance spectra of [(reline + TFE), panel A] and [(glyceline + TFE), panel B] mixtures under ambient conditions. Dashed vertical lines denote shifts, if any, in peak positions.

suggesting that the DES completely disrupts this noncovalent H-bonded interaction within TFE. The −OH stretch representing polymeric TFE structure with extended H-bonding is assigned to the broad band centered at 3363 cm^{-1}.[43−45] This band appears to lose its identity in the (reline + TFE) mixture as the FTIR spectrum of the mixture in this region at x_{TFE} = 0.25 is dominated by the peaks characterizing DES reline. This, in turn, suggests weakening of intermolecular TFE H-bonding interactions as well. In reline, the bands centered at ∼3318 and ∼3190 cm^{-1}, representing H-bonding interactions between ChCl and the HBD urea, show small monotonous hypsochromic shifts of ∼4 and ∼6 cm^{-1}, respectively, upon addition of up to x_{TFE} = 0.25. This may imply slight strengthening of H-bonding between ChCl and urea within the mixture. This proposition is further supported by a significant bathochromic shift of the band characterizing NH_2 rocking from 1164 to 1141 cm^{-1} as up to x_{TFE} = 0.25 is added to reline. A net increase in the strength of H-bonding within the mixture would lead to weakening of NH_2 rocking vibrations within urea. Overall, the FTIR absorbance spectra hints at minimal participation of TFE in H-bonding interactions within the (reline + TFE) mixture.

We have next collected the Raman spectra of neat reline and (reline + TFE) mixtures (Figure 8A). Our neat reline Raman spectral features are the same as those reported earlier.[46] It is pointed-out that Raman bands assigned to antisymmetric and symmetric NH_2, CN, and CO stretching vibrations of urea as well as several vibrational bands assignments of solid ChCl all change to broader bands upon mixing of ChCl with urea to form DES reline. The broadening of the Raman bands of urea and ChCl in reline is attributed to formation of H-bonds between ChCl and urea. Raman spectral features of (reline + TFE) mixtures further corroborate outcomes from FTIR absorbance data. A careful examination of the Raman spectra of the mixtures presented in Figure 8A reveals no significant shifts in any of the Raman bands as TFE is added. However, the presence of a small extent of band broadening upon TFE addition to reline may be argued from the data. This small band

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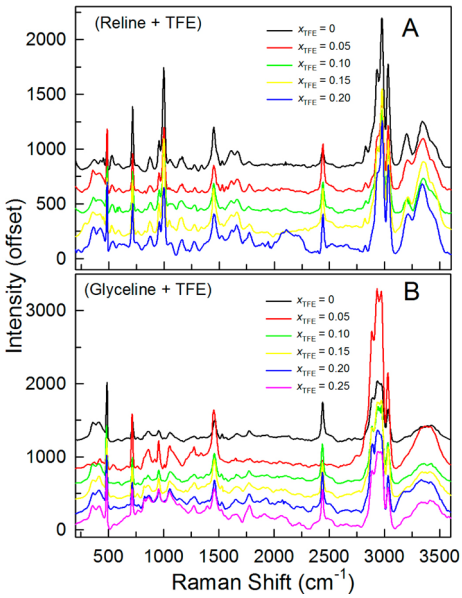


Figure 8. Raman spectra ($\lambda_{excitation}$ = 532 nm) of [(reline + TFE), panel A] and [(glyceline + TFE), panel B] mixtures under ambient conditions.

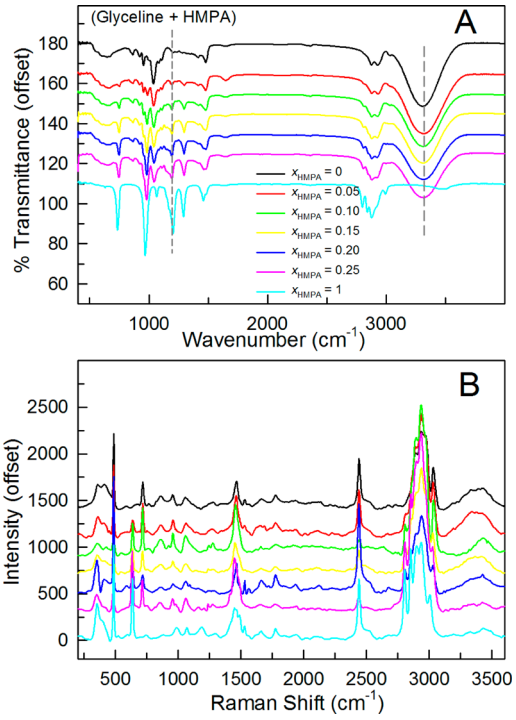


Figure 9. FTIR absorbance (panel A) and Raman (panel B) spectra of (glyceline + HMPA) mixtures under ambient conditions. Dashed vertical lines denote IR assignment shifts, if any, in peak positions.

broadening may hint at the presence, albeit very weak, of reline−TFE interactions.

The band representing H-bonding interaction between ChCl and glycerol (at 3308 cm^{-1}) in glyceline does not exhibit any measurable shift as TFE up to x_{TFE} = 0.25 is added (Figure 7B). However, the presence of H-bonding interactions between HBD TFE and HBA glycerol may not be completely ruled out based on FTIR absorbance data due primarily to the limited sensitivity inherent to the technique. Similar to that in the case of the formation of reline from ChCl and urea, the band broadening for Raman active vibrational transitions of ChCl and glycerol may be suggested when they are mixed in 1:2 mol ratio to form glyceline. Again, the band broadening could be due to the H-bonding interactions between ChCl and glycerol. A careful examination of the Raman spectra of glyceline and TFE-added glyceline (Figure 8B) suggests the absence of band broadening for bands below 2500 cm^{-1} as TFE is added. However, structural smearing to some extent is noticed in the bands present in 2800−3500 cm^{-1} spectral region of glyceline as TFE is added. As the bands in this region usually correspond to vibrations involving −OH groups, the presence of interactions between TFE and glyceline may not be completely ruled out.

FTIR absorbance and Raman spectra of (reline/glyceline + TFE) mixtures (up to x_{TFE} = 0.25) do not exhibit significant shifts in band positions, though subtle broadening in band shapes may be deciphered. Reline and glyceline are constituted of HBDs, urea, and glycerol, respectively, and we believe that the addition of another external HBD cosolvent (TFE up to 0.25 mole fraction) does not result in significant enough changes in the H-bonding of the mixture to become prominently manifested through vibrational band positions of FTIR absorbance and Raman spectra of the mixtures.

HMPA-Added Glyceline Mixtures. As HMPA up to x_{HMPA} = 0.25 is added to glyceline, no significant shift is observed in the band at 3308 cm^{-1} suggesting not much alteration in the H-bonding interactions between ChCl and glycerol (Figure 9A). However, a hypsochromic shift of ∼7 cm^{-1} in the band centered at 1037 cm^{-1} upon x_{HMPA} = 0.25 addition indicating a strengthening in −C−O− stretching vibration suggests involvement of glycerol/cholinium with HMPA through H-bonding. This is further supported by a hypsochromic shift of ∼6 cm^{-1} of the band centered at 1292 cm^{-1} in HMPA representing stretching vibrations of phosphoramide [P−N] bond.[47−49] These outcomes are partly supported by the Raman spectra of (glyceline + HMPA) mixtures (Figure 9B). A careful examination reveals band broadening for the bands in the regions 1400−1500 cm^{-1} and 3100−3600 cm^{-1} characterizing −OH vibrations. Even the bands in the region 2800−3100 cm^{-1} exhibit alteration in band shapes and features. The presence of interactions between added HBA cosolvent HMPA and HBD constituent of the DES glycerol is substantiated from FTIR absorbance and Raman spectroscopic data of the (glyceline + HMPA) mixtures.

Density and Dynamic Viscosity Measurements. To gain further insight into the interactions present, The density and dynamic viscosity of (reline/glyceline + TFE/HMPA) mixtures are measured at ambient conditions. Measurement of density and dynamic viscosity is another way of nonintrusively investigate the interactions present within a complex system.

TFE-Added Reline/Glyceline Mixtures. As the density of neat TFE is higher than that of neat reline and neat glyceline, respectively, the density of the mixture steadily increases as TFE up to x_{TFE} = 0.25 is added to reline/glyceline. To assess the deviation from ideality, if any, excess molar volumes (V^E) of the two mixtures are estimated from the density data (Figure 10). A careful examination of the V^E data presented in Figure 10A reveals interesting outcomes. For both TFE-added reline and glyceline mixtures, the V^E is significantly negative at all mixture compositions until x_{TFE} = 0.25; with the absolute value of V^E being more for the (reline + TFE) mixture as compared to (glyceline + TFE) mixture. The negative V^E, in general, is

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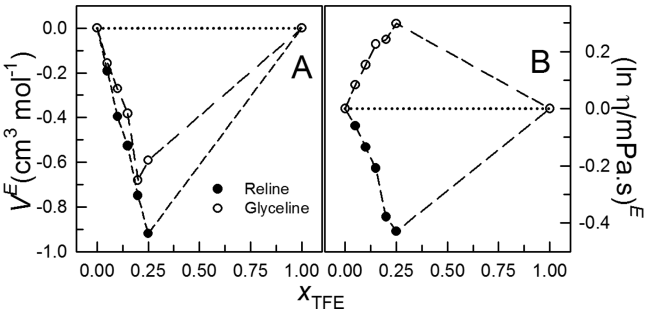


Figure 10. Variation in excess molar volume (V^E, panel A) and excess logarithmic viscosity [$(\ln \eta)^E$, panel B] for TFE-added reline/glyceline mixtures under ambient conditions. The dashed lines are simply to guide the eyes.

ascribed to contraction in volume upon mixing.[50,51] The contraction in volume could be either due to the presence of relatively stronger overall interspecies (possibly between TFE and ChCl) attractive interactions (including H-bonding) as compared to intraspecies (between TFE−TFE and/or reline−reline) attractive interactions within the mixture or it could be due to the interstitial accommodation of TFE within H-bonded network of reline/glyceline.[50,51]

Further insight into this is afforded by the measured dynamic viscosity data of the (reline/glyceline + TFE) mixtures at 25 °C. Dynamic viscosities of reline/glyceline are considerably higher than that of TFE. As expected, dynamic viscosities of both the DESs decrease monotonically as TFE up to x_{TFE} = 0.25 is added. The deviation of measured mixture dynamic viscosities from the ideal additive nature is afforded through the calculation of excess logarithmic dynamic viscosities $(\ln \eta)^E$.[50,51] The $(\ln \eta)^E$ variation with x_{TFE} for (reline/glyceline + TFE) mixtures are presented in Figure 10B. It is interesting to note that while for TFE-added reline $(\ln \eta)^E$ are negative at all compositions investigated for $x_{TFE} \leq 0.25$, for TFE-added glyceline within the same mixture composition, on the other hand, $(\ln \eta)^E$ are all positive. Intensive interspecies attractive interactions (including H-bonding) within the mixture between TFE and the DES would usually lead to positive $(\ln \eta)^E$ as mixture viscosity becomes higher than the calculated ideal viscosity. On the contrary, the interstitial accommodation of TFE within the H-bonded network of the DES may lead to negative $(\ln \eta)^E$ as, in this case, the mixture viscosity would be less than that calculated assuming ideal additive behavior. The intraspecies interactions are usually disrupted in this case and the interspecies interactions are usually not too significant.

On the basis of V^E and $(\ln \eta)^E$ data of the TFE-added DESs, it is difficult to propose the presence of stronger H-bond interactions between reline and TFE within the mixture to be the prominent interaction as $(\ln \eta)^E$ values are negative, although the presence of such H-bonding interactions may not be completely ruled out. Interstitial accommodation of TFE within H-bonded reline network along with possible minor contribution from H-bonding and other interactions between the unlike components (i.e., ChCl and TFE) within the mixture may result in volume contraction (and negative V^E) as well as lowered dynamic viscosity than expected [i.e., negative $(\ln \eta)^E$]. For TFE-added glyceline, the V^E and $(\ln \eta)^E$ data in combination hints at the presence of relative stronger H-bonding between TFE and glyceline. This would lead to contraction in volume (and hence negative V^E) and higher than expected bulk dynamic viscosity [and hence positive $(\ln \eta)^E$] as TFE is added to glyceline. The more efficient H-bonding ability of glycerol over urea as glycerol possesses three alkyl −OH groups as opposed to two $-NH_2$ groups on C═O functionality in urea is tentatively attributed to be the reason.

HMPA-Added Glyceline Mixtures. The density of neat HMPA is smaller than that of neat glyceline at 25 °C. As HMPA up to x_{HMPA} = 0.25 is added to glyceline, the density steadily decreases. V^E values are estimated from the measured mixture densities and the densities of the neat HMPA and neat glyceline (Figure 11A). Surprisingly, in complete contrast to

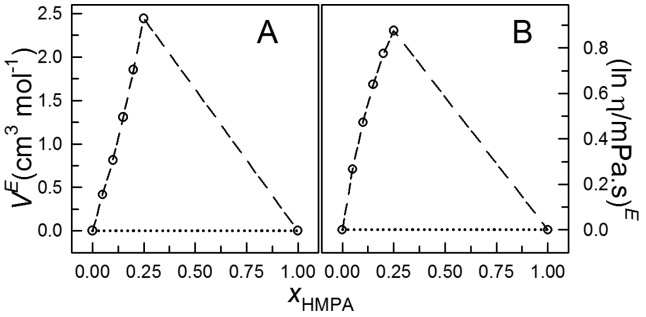


Figure 11. Variation in excess molar volume (V^E, panel A) and excess logarithmic viscosity [$(\ln \eta)^E$, panel B] for HMPA-added glyceline mixtures under ambient conditions. The dashed lines are simply to guide the eyes.

that observed for (reline/glyceline + TFE) mixtures, the V^E for the (glyceline + HMPA) mixture in the same composition regime ($x_{HMPA} \leq 0.25$) are all positive with much higher absolute V^E values. A positive V^E here implies expansion in volume as HMPA is mixed with glyceline. This hints at the disruption of overall attractive interactions (prominently H-bonding and electrostatic) present within glyceline (between ChCl and glycerol) as HMPA is added. Dynamic viscosity of neat HMPA is much smaller than that of glyceline. However, surprisingly, the dynamic viscosity of glyceline increases initially before decreasing as HMPA up to x_{HMPA} = 0.25 is added (for x_{HMPA} = 0.05, 0.10, and 0.15, the dynamic viscosity of the mixture is higher than that of neat glyceline). Subsequently, this leads to significantly higher positive $(\ln \eta)^E$ values for HMPA-added glyceline mixtures as compared to those for TFE-added glyceline at the same compositions (Figure 11B). The presence of increased overall net repulsive interactions within the (glyceline + HMPA) mixtures (possibly involving HBAs, HMPA, and ChCl) is proposed to explain significantly positive V^E and $(\ln \eta)^E$.

The dynamic viscosity of neat glycerol is ∼942 mPa·s at 25 °C.[50] It decreases to ∼191 mPa·s as ChCl is mixed with it in 2:1 molar ratio to form DES glyceline. Although the reason for DES formation, in major part, is proposed to be the formation of H-bonding between Cl^- of ChCl and HBD glycerol −OH groups,[2−6] the overall change in H-bonding interactions within the system gives rise to decreased viscosity. The addition of HMPA may lead to formation of H-bonds between HBA HMPA and HBD glycerol which, in turn, disrupts the interactions between ChCl and glycerol in the system. The HMPA-glycerol H-bonding is more similar in nature to glycerol−glycerol H-bonding than ChCl−glycerol H-bonding. This leads to increased viscosity of the mixture and significantly positive $(\ln \eta)^E$. However, disruption in the overall net interactions within the system (between ChCl−glycerol and HMPA−HMPA), in turn, leads to positive V^E.

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CONCLUSIONS

The significant lowering of the melting points to form DESs reline and glyceline is attributed to the H-bonding interaction(s) between the salt ChCl and HB donors urea and glycerol, respectively. Addition of a cosolvent TFE of very high HB donating acidity and no HB accepting basicity or a cosolvent HMPA of very high HB accepting basicity and no HB donating acidity to reline and glyceline, respectively, appears to disturb the ChCl−urea/glycerol equilibria within the DES. Subsequently, only up to 0.25 mole fraction of TFE is miscible in both reline and glyceline as above this concentration of TFE within the system, the HB donors urea and glycerol, respectively, appear to separate out. Similar behavior is observed for HMPA in glyceline; however, this cosolvent is found to be completely immiscible with reline perhaps due to the formation of an insoluble HB acceptor HMPA−HB donor urea complex. Responses of the solvatochromic absorbance and fluorescence probes are found to depend on the identity of the probe. The responses of the same probes within TFE-added prototypical ionic liquid 1-butyl-3-methylimidazolium hexafluorophosphate are vastly different as no "hyperpolarity" is observed for (reline/glyceline + TFE) mixtures possibly hinting at the absence of strong interspecies H-bonding interactions within TFE-added DESs. FTIR absorbance and Raman spectroscopic measurements, in part, corroborate this observation. A combination of V^E and $(\ln \eta)^E$ of the mixtures hint at possible interstitial accommodation of TFE within H-bonded network of reline, however, it appears to form H-bonds with the constituents of glyceline. Positive V^E and positive $(\ln \eta)^E$ of (glyceline + HMPA) mixtures suggest an increase in overall net repulsive interactions within the systems. The overall outcomes of the investigation suggest that the fascinating and attractive features of DESs can be considerably expanded by mixing the DESs with suitable cosolvents, opening their versatility beyond the pure materials.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcb.7b01724.

Miscibility of HBD/HBA cosolvents TFE/HMPA with DESs reline and glyceline, respectively, under ambient conditions (Figure S1)(PDF)

AUTHOR INFORMATION

Corresponding Author

*E-mail: sipandey@chemistry.iitd.ac.in. Tel.: +91-11-26596503. Fax: +91-11-26581102.

ORCID

Siddharth Pandey: 0000-0003-3175-5353

Notes

The authors declare no competing financial interest.

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